Database Project - Railway Transport and Reservation Management System

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1. Objective

The goal here is to "design and implement" the database for an identified application. This system acts as an electronic marketplace to manage railway transport and reservations, similar to a simplified IRCTC system.

The database will efficiently manage trains, stations, train routes, passengers, and train classes (1A, 2A, 3A, Sleeper, etc.). It will also track coaches, seats, ticket bookings, PNR generation, class-wise waiting lists, ticket cancellations, payments, and complex administrative reports. The objective is to ensure that all required data are included in the proposed database schema.

2. Application Users & Use Cases

Users interact with the system through specific activities. We describe the use cases and queries for building the application.

Use Cases - Guest (Unregistered User)

These are the users who just land on the website and just explore around without registering and logging in.

- **Search Trains:** Search for available trains between a source and destination.
- **View Train Routes:** View the sequence of stations and timings for specific trains.

- **Check Seat Availability:** View real-time seat availability categorized by train class.

Use Cases - Passenger

Passengers (let us also call them customers) also require registration.

- **Register:** Create an account in the system.
- **Modify Details:** Update profile information (name, contact, etc.).
- **Search Trains:** Find trains by date and route.
- **Book Tickets:** Choose a train class and submit a reservation.
- **View Booking History:** Access past and upcoming journeys.
- **Check PNR Status:** Track the current confirmation status of a ticket.
- **Cancel Tickets:** Cancel valid tickets prior to chart preparation.
- **Give Feedback:** Rate travel experiences and leave review comments.

Use Cases - Administrator

Administrators manage the backend operations and master data.

- **Manage Trains & Stations:** Add new trains and manage station details.
- **Define Train Routes:** Map out the exact path and schedule of every train.
- **Assign Coaches:** Map specific coach types to trains.
- **Manage Bookings & Waiting Lists:** Oversee system-wide reservations and waiting list movements.
- **Generate Reports:** Extract operational and financial summaries for decision-making.

3. Train and Station Management

The database accurately models the physical railway network.

- **Trains:** Each train is stored with a unique train number, train name, source station, and destination station.
- **Stations:** Stations are recorded with a unique station code, station name, and city.
- **Routes:** Because each train travels through multiple stations, a train route table maps the sequence of stops, arrival times, and departure times.

4. Train Classes

Different travel classes dictate the pricing and coach layouts. Supported classes include:

- First AC (1A)
- Second AC (2A)
- Third AC (3A)

- Sleeper (SL)
- Chair Car (CC)
- General (GEN)

The database enforces that each class has a different price, uses different specific coaches, and possesses a distinct total seat count.

5. Seat Reservation System

When a passenger books a ticket, they select the train, class, and travel date.

The system queries the database for seat availability. If seats are available, a specific seat is assigned (linked through the coach and seat tables), the booking is confirmed, and a unique PNR is generated.

6. Class-Wise Waiting List

To handle high demand, the database utilizes a waiting list logic.

If all seats in a requested class are full, the passenger is placed on a waiting list specific to that class. For example, on Train 12931, one passenger might be "Sleeper WL3" while another is "3AC WL1". When confirmed tickets are cancelled, the system automatically updates the waiting list, promoting passengers to confirmed status.

7. Ticket Cancellation and Refund

When a passenger cancels a ticket, the database updates the booking status.

The canceled seat immediately becomes available, triggering the waiting list logic to confirm the next passenger. Necessary data for cancellation of un-dispatched orders and status of refund of such orders be also be accounted for.

8. Payment System

Payments are often done by external payment gateways therefore let us keep the payment process out of the scope of our database. Only the things that we should store are the Payment Service name and payment acknowledgment reference number. The system will also record the booking ID, payment method, amount, and payment status.

9. Reports and Queries

The database must answer complex questions for system users. Key reports include:

- Trains operating between two specific stations.
- Seat availability categorized by class.
- Passenger booking history and itineraries.
- Passenger manifest for a specific train and date.
- List of waiting list passengers.
- Train occupancy reports.
- Revenue generated by train and by class.
- Cancellation and refund statistics.

10.Key Questions the Database Must Answer

- **For Guests & Passengers**

  What available trains operate between a specific source and destination station on a given date?

  What is the exact sequence of stations, arrival times, and departure times for a specific train route?

  What is the real-time seat availability categorized by specific train classes (e.g., 1A, Sleeper, General)?

  What is the current confirmation and PNR status of my booked ticket?

  What is the payment status, payment method, and acknowledgment reference number for my booking?

  What is my complete booking history, including past and upcoming itineraries?

- **For Administrators**

  What is the passenger manifest for a specific train and travel date?

  Which passengers are currently on the class-wise waiting list for a specific train (e.g., "Sleeper WL3")?

What are the current train occupancy rates across different travel classes?

How much total revenue has been generated, categorized by specific trains and by train classes?

What are the statistics for ticket cancellations and the refund status of un-dispatched orders?

Which specific coach types are mapped and assigned to a particular train?

Are there any canceled tickets that need to trigger the waiting list logic to confirm the next passenger?

11. Conclusion

You should be conceptualizing a real application with a real user. This database accurately models a complex railway reservation system, supporting multiple entities, intricate relationships, and dynamic reporting queries suitable for a comprehensive DBMS project